ROLE OF THE CARDIAC CYCLE IN THE PERCEPTION OF REALITY AND ILLUSION

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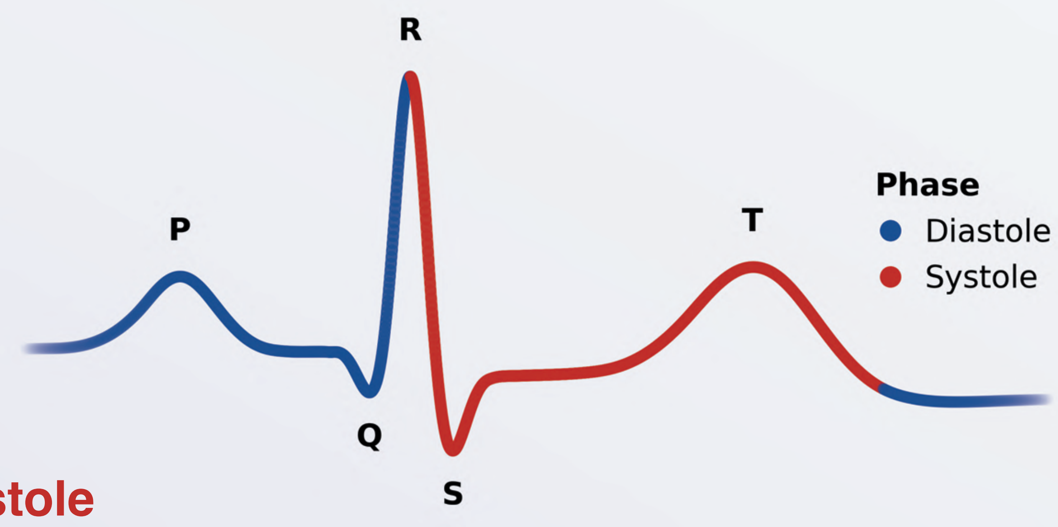
References and Acknowledgements

1. BACKGROUND

Over one heartbeat, the heart completes the **cardiac cycle**, comprising **diastole** and **systole**

Signals from the heart seem to play an important role in perception and cognition (Azzalini et al., 2019)

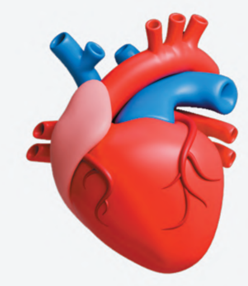
In particular, signals transmitted to the brain during **systole** may reduce the accuracy of perception (Skora et al., 2022)



ECG trace over one heartbeat

To understand whether our sense of "reality" may be affected by cardiac signals, we can measure how strongly people perceive **visual illusions** (VIs) to quantify one's "**VI sensitivity**". Possessing a higher VI sensitivity means that an individual views a stimulus as having a subjectively stronger illusory effect

2. AIMS AND OBJECTIVES



Aim: To investigate whether the cardiac cycle is related to the perception of VIs

Hypothesis: In the "Illusion Game" task, VI sensitivity will be stronger (more error-prone) when participants view VIs during **systole** (relative to **diastole**)

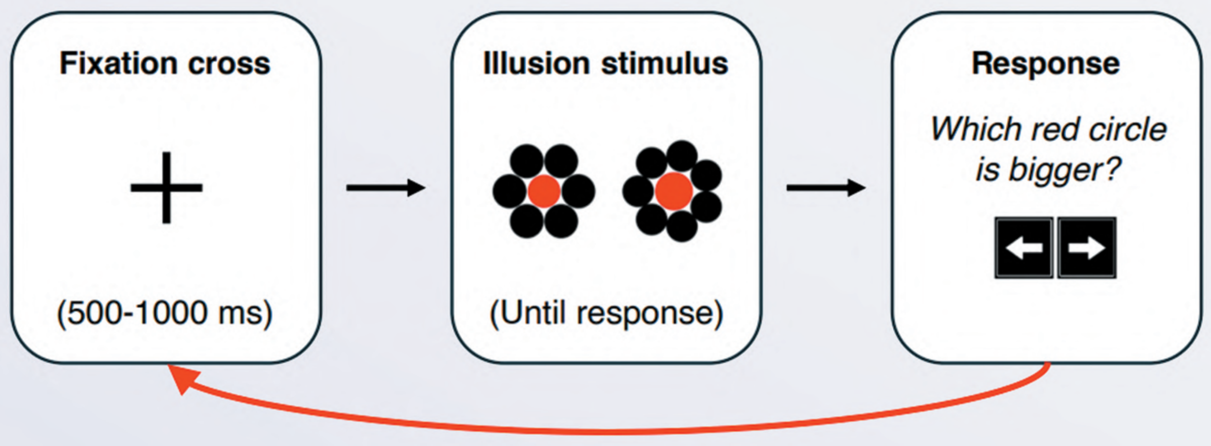
3. PARTICIPANTS

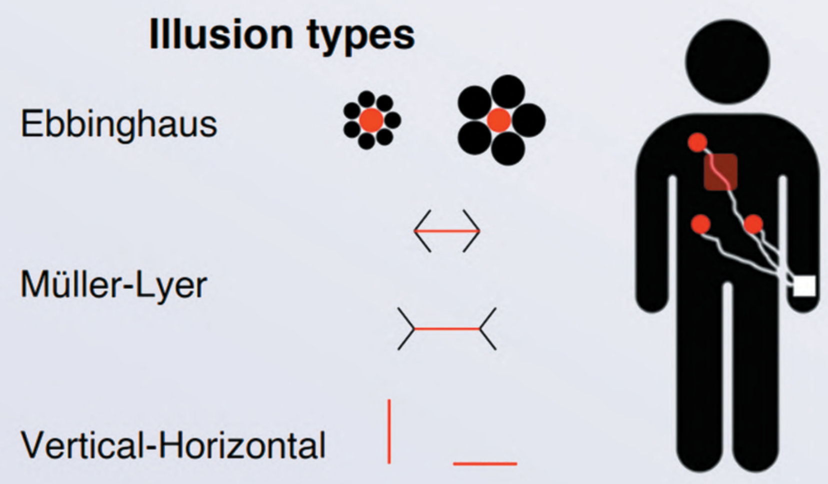
33 participants (M_{Age} = 25.2, SD = 8.7, range: [18, 66], 57.6% ♀) were recruited at the University of Sussex

4. METHODS

Participants completed the **Illusion Game** (Makowski et al., 2023) whilst their cardiac activity was measured using a 3-lead **electrocardiogram** (ECG)

- In this speed-accuracy task, participants made perceptual judgements across three illusion types: **Ebbinghaus**, **Müller-Lyer**, and **Vertical-Horizontal** (e.g., for the Ebbinghaus illusion, participants answered "Which red circle is bigger?" by making correct/incorrect arrow key responses)
- Over 480 trials, **task difficulty** (core illusion elements) and **illusion strength** (illusory context) of the stimuli were manipulated independently and continuously
- Trials were either **congruent** or **incongruent**, i.e., influencing judgements towards or against the correct alternative, respectively. Only **incongruent** trials are included in the analysis
- To quantify VI sensitivity, we can compute a participant's average **error rate** (↑ VI Sens. = ↑ Errors)

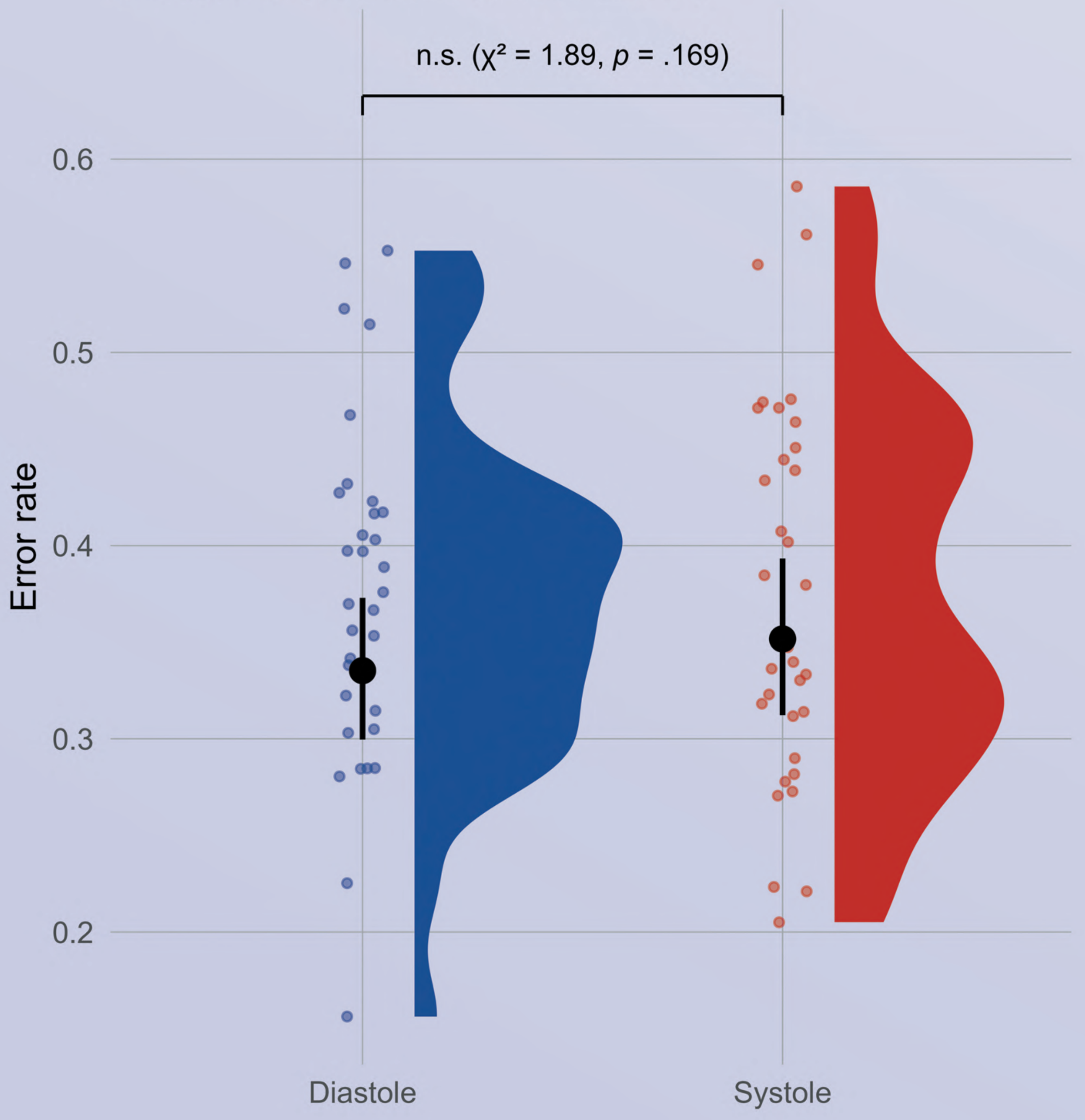




5. RESULTS

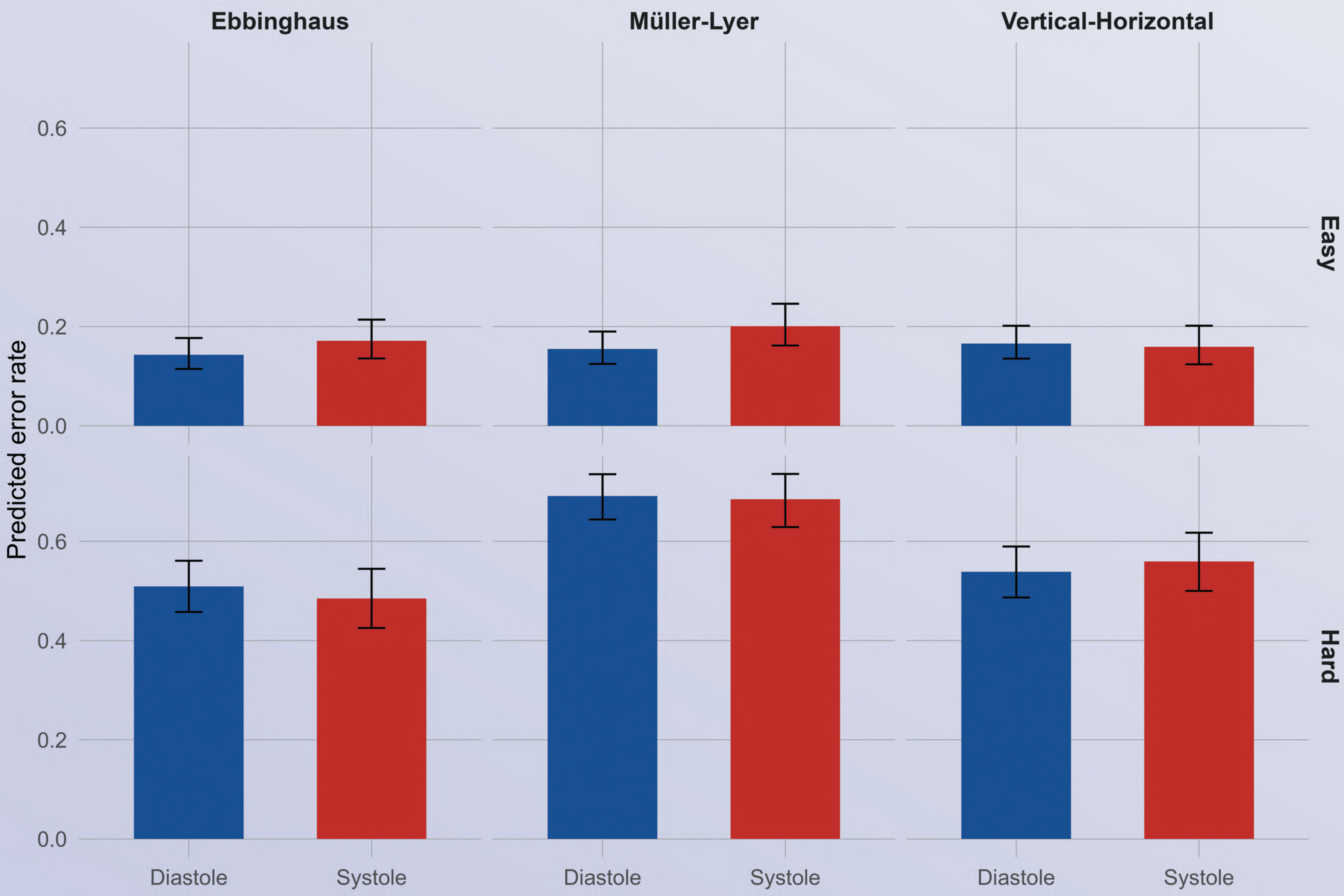
Overall effect of cardiac phase on error rate

Averaged across illusion type and difficulty



Cardiac phase effect by illusion type and difficulty

Cardiac×Illusion: n.s. (χ^2 = 3.07, p = .216) | Cardiac×Difficulty: n.s. (χ^2 = 2.56, p = .109) | 3-way: n.s. (χ^2 = 3.81, p = .149)



Statistical modelling (logistic regression) found **no** evidence that cardiac phase influenced **error rate**, either overall, $\chi^2(1) = 1.89$, $p = .169$, or in interaction with illusion type, $\chi^2(2) = 3.07$, $p = .216$, difficulty, $\chi^2(1) = 2.56$, $p = .109$, or the three-way combination of phase, illusion type, and difficulty, $\chi^2(2) = 3.81$, $p = .149$

6A. DISCUSSION

Overall, the results **fail** to support the hypothesis that VI sensitivity is stronger in **systole** (versus **diastole**)

- At face value, this would suggest that VI sensitivity may **not** be modulated by cardiac phase effects
- **However**, it is important to recognise that this sample only consisted of 33 participants, which may constitute insufficient statistical power to detect such a small effect of the cardiac cycle
- **Additionally**, cardiac phase was coded as categorical (**diastole**/**systole**) rather than continuous (**0-100%**), potentially masking variability in cardiac phase dynamics on VI sensitivity

6B. FUTURE DIRECTIONS

We aim to carry out the following:

- Recruit 100 total participants
- Develop more complex computational models to simulate decision processes underlying VIs
- Include measures pertaining to interoception, cognitive control, and cardiac cycle % completed